Pierre Marza

Assistant Professor, CentraleSupélec, Université Paris-Saclay
Deep Learning, Computer Vision, Medical Imaging, Digital Pathology
Webpage | Google Scholar | GitHub | Linkedin

Experience

July, 2026 – Present — ***Assistant Professor (Maître de Conférences)***, *CentraleSupélec (Cancer Data Science Unit, MICS)*, Gif-Sur-Yvette, France.

January, 2025 – July, 2026 — ***Postdoctoral Researcher***, *CentraleSupélec (MICS)*, Gif-Sur-Yvette, France.
- Benchmarking Digital Pathology Foundation Models
- Latent space augmentations
- Causal attention regularization
- Aligning Digital Pathology and Spatial Transcriptomics
- Pathology slide registration
- Agents for histopathology

Collaborators: **Maria Vakalopoulou** (→), **Stergios Christodoulidis** (→)

June, 2022 – Oct., 2022 — ***Research intern - Embodied AI***, *Meta AI (FAIR)*, Menlo Park, California, US.
- Embodied active learning
- Semantic Neural Radiance Fields

Advisors: **Devendra Singh Chaplot** (→), **Dhruv Batra** (→)

Feb., 2020 – Aug., 2020 — ***Research intern - Visual Question Answering***, *ORANGE LABS*, Rennes, France.
- Object Detection in images - Attention mechanisms
- Guiding detection of salient regions with textual information

Advisors: **Corentin Kervadec** (→), **Grigory Antipov** (→), **Moez Baccouche** (→), **Christian Wolf** (→)

May, 2019 – Dec., 2019 — ***Research Intern - Computer Vision***, *HUAWEI Noah's Ark Lab*, London, UK.
- Research work on Image Quality Enhancement (deblurring, denoising, demosaicing) with Deep Learning
- Neural Architecture Search (NAS)

Advisors : **Sean Moran** (→), **Greg Slabaugh** (→)

June, 2018 – August, 2018 — ***Python development - Neural Networks***, *SOGETI High Tech*, Lyon, France.
Chatbot - Recurrent Neural Networks (Seq2Seq, LSTM)

Education

2020 – 2024 — **PhD, Embodied AI**, *LIRIS/CITI*, INSA Lyon, France.
- Embodied AI, Computer Vision, Deep Learning, Reinforcement Learning

Advisors : **Christian Wolf** (→), **Laetitia Matignon** (→), **Olivier Simonin** (→)

2017–2020 : **Master in Computer Science**, *INSA Lyon*, France.
- Research & Development Specialization
- Exchange Semester at KTH, Stockholm, Sweden (Machine Learning, Deep Learning, Reinforcement Learning)

2015–2017 : **Preparatory classes (Scientific common core)**, *INSA Lyon*, France.

Papers/Patents

2026 — Ruoyu Xue, Suryakant Singh, Souradeep Chakraborty, **Pierre Marza**, Oksana Yaskiv, Constantin Friedman, Natallia Sheuka, Paul Friedman, Bharat Ramlal, Beatrice Knudsen, et al. Pathologist attention–aligned report generation for prostate histopathology. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026.

2026 Leo Fillioux, Omprakash Chakraborty, Quentin Gopée, **Pierre Marza**, Paul-Henry Cournède, Stergios Christodoulidis, Maria Vakalopoulou, Ismail Ben Ayed, and Jose Dolz. Information maximization for long-tailed semi-supervised domain generalization. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026.

2026 Imane Chraki, **Pierre Marza**, Stergios Christodoulidis, and Maria Vakalopoulou. Counterfactual intervention in attention multiple instance learning for digital pathology. *Medical Imaging with Deep Learning (MIDL)*, 2026.

2026 Imane Chraki, **Pierre Marza**, Stergios Christodoulidis, and Maria Vakalopoulou. Car-mil: Counterfactual attention regularization for multiple instance learning. *European Conference on Computer Vision (ECCV)*, 2026.

2026 Sofiène Boutaj*, **Pierre Marza***, Varun Belagali, Dimitris Samaras, Maria Vakalopoulou, and Stergios Christodoulidis. Scaling multi-teacher distillation for digital pathology. *Neural Information Processing Systems (NeurIPS)*, 2026.

2026 Sofiène Boutaj, Leo Fillioux, Maria Vakalopoulou, Stergios Christodoulidis°, and **Pierre Marza°**. From patches to patients: A study of the tile-to-slide performance transferability in digital pathology. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026.

2026 Varun Belagali, **Pierre Marza**, Srikar Yellapragada, Zilinghan Li, Tarak Nath Nandi, Ravi K Madduri, Joel Saltz, Stergios Christodoulidis, Maria Vakalopoulou, and Dimitris Samaras. Cdg-mae: Learning correspondences from diffusion generated views. *Transactions on Machine Learning Research (TMLR)*, 2026.

2026 Varun Belagali*, Saarthak Kapse*, **Pierre Marza°**, Srijan Das°, Zilinghan Li, Sofiène Boutaj, Pushpak Pati, Srikar Yellapragada, Tarak Nath Nandi, Ravi K Madduri, Joel Saltz, Prateek Prasanna, Stergios Christodoulidis, Maria Vakalopoulou, and Dimitris Samaras. Ticon: A slide-level tile contextualizer for histopathology representation learning. *Computer Vision and Pattern Recognition (CVPR) Findings*, 2026.

2025 **Pierre Marza**, Leo Fillioux*, Sofiène Boutaj*, Kunal Mahatha, Christian Desrosiers, Pablo Piantanida, Jose Dolz, Stergios Christodoulidis°, and Maria Vakalopoulou°. Thunder: Tile-level histopathology image understanding benchmark. *Neural Information Processing Systems (NeurIPS) D&B Track (Spotlight)*, 2025.

2025 Kunal Mahatha, Ali Bahri, **Pierre Marza**, Sahar Dastani, Maria Vakalopoulou, Stergios Christodoulidis, Jose Dolz, and Christian Desrosiers. Octopus: Enhancing the spatial-awareness of vision ssms with multi-dimensional scans and traversal selection. *arXiv*, 2025.

2025 Sofiène Boutaj, Marin Scalbert, **Pierre Marza**, Florent Couzinie-Devy, Maria Vakalopoulou, and Stergios Christodoulidis. Controllable latent space augmentation for digital pathology. In *International Conference on Computer Vision (ICCV)*, 2025.

2024 **Pierre Marza**, Laetitia Matignon, Olivier Simonin, and Christian Wolf. Task-conditioned adaptation of visual features in multi-task policy learning. *Computer Vision and Pattern Recognition (CVPR)*, 2024.

2024 **Pierre Marza**, Laetitia Matignon, Olivier Simonin, Dhruv Batra, Christian Wolf, and Devendra Singh Chaplot. Autonerf: Training implicit scene representations with autonomous agents. *International Conference on Intelligent Robots and Systems (IROS)*, 2024.

2023 **Pierre Marza**, Laetitia Matignon, Olivier Simonin, and Christian Wolf. Multi-object navigation with dynamically learned neural implicit representations. *International Conference on Computer Vision (ICCV)*, 2023.

2022 **Pierre Marza**, Laëtitia Matignon, Olivier Simonin, and Christian Wolf. Teaching agents how to map: Spatial reasoning for multi-object navigation. *International Conference on Intelligent Robots and Systems (IROS)*, 2022.

2022 **Pierre Marza**, Corentin Kervadec, Grigory Antipov, Moez Baccouche, and Christian Wolf. An experimental study of the vision-bottleneck in vqa. *arXiv*, 2022.

2021 Sean Moran, **Pierre Marza**, Steven McDonagh, Sarah Parisot, and Gregory Slabaugh. A device and method for image processing. *WO Patent*, 2021.

2020 Sean Moran, **Pierre Marza**, Steven McDonagh, Sarah Parisot, and Gregory Slabaugh. Deeplpf: Deep local parametric filters for image enhancement. *Computer Vision and Pattern Recognition (CVPR)*, 2020.

Open-source contributions

2025 **THUNDER**, A tile-level histopathology image understanding benchmark.
https://mics-lab.github.io/thunder/

2024 **Task-conditioned adaptation**, Code for CVPR 2024 paper *Task-conditioned adaptation of visual features in multi-task policy learning*.
https://github.com/PierreMarza/task_conditioned_adaptation

2024 **AutoNeRF**, Code for IROS 2024 paper *AutoNeRF: Training Implicit Scene Representations with Autonomous Agents*.
https://github.com/PierreMarza/autonerf

2023 **Dynamic Implicit Representations**, Code for ICCV 2023 paper *Multi-Object Navigation with dynamically learned neural implicit representations*.
https://github.com/PierreMarza/dynamic_implicit_representations

2022 **Teaching Agents How to Map**, Code for IROS 2022 paper *Teaching Agents how to Map: Spatial Reasoning for Multi-Object Navigation*.
https://github.com/PierreMarza/teaching_agents_how_to_map

2020 **DeepLPF**, Code for CVPR 2020 paper *Deep Local Parametric Filters for Image Enhancement*.
https://github.com/sjmoran/deeplpf-image-enhancement

Teaching

2025–2026 **Maths for oncology projects**, DU Mathématiques appliquées au Cancer, 13 students (clinicians, biologists, bio-engineers).
Co-evaluating 8 groups of 2 students on various topics related to the application of AI to oncology.

2025–2026 **Python and maths refresher**, DU Mathématiques appliquées au Cancer (14h), 13 students (clinicians, biologists, bio-engineers).
Introduction to Python programming and basics of mathematical functions.

2025–2026 **Reinforcement Learning MSc AI**, CentraleSupélec (10.5h), Master-level, 91 students.
Reinforcement Learning labs (Exploration/exploitation, Dynamic Programming, Monte Carlo Methods, Value Function Approximation, Policy Gradient) and project.

2025–2026 **Reinforcement Learning MSc DSBA**, CentraleSupélec (12h), Master-level, 50 students.
Reinforcement Learning labs (Exploration/exploitation, Dynamic Programming, Monte Carlo Methods, Value Function Approximation, Policy Gradient), one lecture (Value Function Approximation) and a project.

2025–2026 **MSc AI projects**, CentraleSupélec, Master-level, 5 students.
Advising two groups of students: Adapting the invariance of self-supervised learning models and Reinforcement Learning for Digital Pathology

2024–2025 **AI Summer School**, CentraleSupélec (9h), Bachelor-level, 11 students.
Deep Learning Lecture, Deep Learning Lab, Reinforcement Learning Lecture

2024–2025 **AI Challenge jury**, CentraleSupélec (3.5h), Master-level.
Part of the evaluation jury for a student project about Computer Vision and AI.

2024–2025 **Reinforcement Learning MSc AI**, CentraleSupélec (12h), Master-level, 71 students.
Reinforcement Learning labs (Exploration/exploitation, Dynamic Programming, Monte Carlo Methods, Value Function Approximation, Policy Gradient) and project.

2024–2025 **Reinforcement Learning MSc DSBA**, CentraleSupélec (12h), Master-level, 26 students.
Reinforcement Learning labs (Exploration/exploitation, Dynamic Programming, Monte Carlo Methods, Value Function Approximation, Policy Gradient) and project.

2023–2024 **Artificial Intelligence and Data Analysis**, Université Lyon 1 (12h), Bachelor-level, 16 students.
Lectures (CNN, RNN), labs (CNNs) and a project (image classification, RL).

2022–2023 **Introduction to Deep Learning**, EPITA Lyon (12h), Master-level, 25 students.
Lectures (CNN, RNN, Transformer), labs (CNN, RNN, Transformer) and a project (rigorous comparison of CNNs, RNNs and Transformers on a movie review sentiment classification problem).

2021–2022 **Deep Learning and Differentiable Programming**, INSA Lyon (8h), Master-level, 21 students.
Deep Learning Lab, Deep Reinforcement Learning Lab, Semi-supervised Classification project.

Reviewing

2026 **ICLR, CVPR, MedIA, MICCAI, NeurIPS (Top reviewer)**.

2025 **ICCV, NeurIPS (Top reviewer)**.

2024 **ICLR, ECCV (Outstanding reviewer), IROS**.

2023 **ICLR, ICCV, NeurIPS**.

2022 **ICML (Outstanding reviewer)**.

2021-2023 **TPAMI**.